Negative acceleration does not always mean slowing

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Average acceleration / ES: aceleración media / PT: aceleração média

Change in velocity divided by elapsed time.

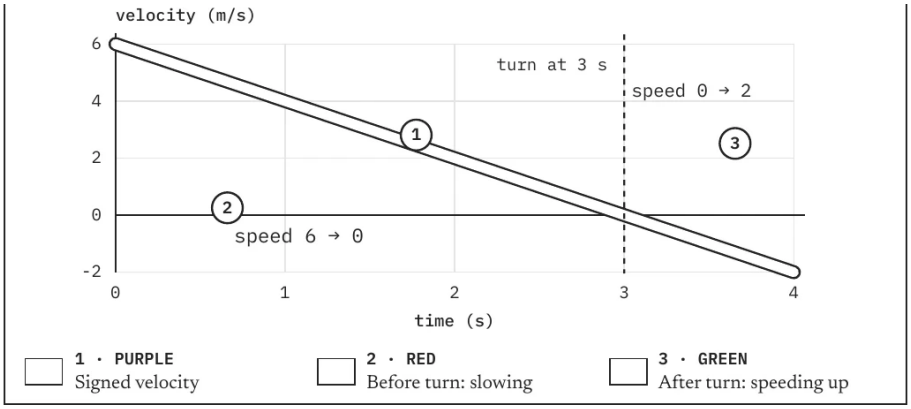
Also on your note page: Speeding up, Slowing down, Reversal.

B READ THE RELATIONSHIP

Average acceleration = change in velocity ÷ time

As you read, find: before · after · change.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart's velocity changes uniformly from +6 m/s to −2 m/s in 4 s . Find its acceleration and the time it reverses.

Given $v_i = +6$ m/s $v_f = -2$ m/s $\Delta t = 4$ s

Rule $a = \Delta v / \Delta t$. Turn: $0 = v_i + at$.

Work $a = (-2 - 6)/4$ $0 = 6 - 2t$ $t = 3$ s

Answer $a = -2$ m/s² Slows for 3 s , then speeds up to the left.

E SAVE FOR THE FINAL MEETING

A change in XJ's velocity requires an acceleration term in the forecast. Its sign alone does not show whether the asteroid is speeding up; compare the directions of velocity and acceleration.

Read the physics: Average acceleration

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Compare velocity before and after. A change in speed or a turn is a change in velocity. Divide the signed change in velocity by the time it took.

02 HOW IT WORKS

Read m/s² as a change in meters per second each second. Look at dot spacing at equal times: growing gaps mean faster motion; shrinking gaps mean slower motion.

03 WHAT TO CHECK

Compare velocity and acceleration. In a straight line, matching directions mean speed grows. Opposite directions mean speed falls until the object stops or reverses.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart's velocity changes uniformly from +6 m/s to −2 m/s in 4 s . Find its acceleration and the time it reverses. a = −2 m/s² Slows for 3 s , then speeds up to the left.

Model check: Average acceleration = change in velocity ÷ time.

05 CONNECT TO XJ

A change in XJ's velocity requires an acceleration term in the forecast. Its sign alone does not show whether the asteroid is speeding up; compare the directions of velocity and acceleration.

SAY IT BEFORE YOU WRITE

Can acceleration be negative while speed increases? The velocity changes from ___ to ___. Each second it changes by ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Average acceleration / ES: aceleración media / PT: aceleração média

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, average acceleration means …	What do you compare at the start and end of the time interval? I find the change in ___ and divide it by ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Draw the before and after velocity arrows (m/s). Mark the change in velocity (m/s). Divide by time to label the acceleration (m/s²).

Labels you can use: before · after · change.

Before	After

Tell a partner: The velocity changes from ___ to ___. Each second it changes by ___.

On assignment: Average acceleration

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two signed XJ velocities are –2 and –6 km/s, two seconds apart. The control room must choose its speed alert.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Telemetry engineer**

Send the average acceleration with its sign and unit. Sketch before and after velocity arrows so the shift lead can confirm the alert.

☐ **Replay commentator**

Narrate what changed over two seconds. Have your producer show which arrow proves whether XJ sped up or slowed down.

Word bank: before · after · change.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”

Editor’s check: Explain why two readings give an average, not the acceleration at every instant.